

Cheng-Yen (Wesley) Hsieh

chengyeh@alumni.cmu.edu | (412) 996-5387 | [in chengyen-hsieh](https://github.com/chengyen-hsieh) | [WesleyHsieh0806](https://www.wesleyhsieh0806.com) | [Personal Website](#)

RESEARCH FOCUS

I build **large-scale generative foundation models** at the intersection of multimodal modeling and scalable training infrastructure. My work spans unified architectures that jointly reason over discrete and continuous modalities, using flow matching, discrete diffusion, and autoregressive frameworks, with applications in **biomolecular design** (proteins, DNA, RNA) and **computer vision** (video generation, vision-language models). I pair modeling contributions with hands-on *pre-training* and *mid-training* infrastructure work, enabling reliable training at scale.

EDUCATION

Carnegie Mellon University

M.S. in Computer Vision (MSCV), School of Computer Science

Pittsburgh, PA

Dec. 2023

National Taiwan University.

B.S. in Electrical Engineering

Taipei, Taiwan

Feb. 2022

WORK EXPERIENCE

ByteDance Seed

Senior Research Scientist

San Jose, CA

Mar. 2024 — Present

Protein Autoregressive Modeling via Multiscale Structure Generation ([PAR](#)) *ICML 2026 Oral (Top 0.7%)*

- Developed **PAR**, the **first** multi-scale autoregressive model for protein backbone generation, designing a multi-scale AR Transformer with a flow-based decoder to directly model Ca atoms, eliminating fidelity loss from structure tokenization.
- Mitigated **exposure bias** via noisy context learning and scheduled sampling; Achieved the state-of-the-art FPSD 161.0 on PDB benchmark, scaling effectively with compute.
- Showed strong zero-shot generalization, reflecting the versatility of AR language models in protein design; Obtained 2.5× sampling speedup over diffusion baselines via multi-scale orchestration of SDE/ODE sampling.

Diffusion-Based Biomolecular Foundation Model – AI for Science

- Building a **700M–3B** unified diffusion model for multi-task understanding and generative design across proteins, DNA, and RNA, supporting binder design, structure prediction, motif scaffolding, and inverse folding.
- Architected the *pre-training* and *mid-training* pipeline for multimodal, multi-task learning at scale; curated an **18M-sample dataset** spanning single-chain proteins, protein complexes, and biomolecular complexes.

Multimodal Diffusion Protein Language Model ([DPLM](#), [DPLM-2.1](#)) *ICML 2025 Spotlight (Top 2.6%)*

- Developed a multimodal protein foundation model that jointly generates protein sequences and 3D structures via **discrete diffusion** protein language model.
- Introduced **bitwise discrete modeling**, a hybrid approach for data-space modeling, representation alignment and geometry-aware architectural designs that significantly improve structural generation and sampling diversity.
- Leveraged multimer data to advance the structural modeling for both single and multi-chain proteins.
- Improve the folding RMSD from **5.52 to 2.36** on the PDB date dataset, outperforming 3B specialized folding baselines using a multimodal PLM with only 650M parameters.

3D-Consistent Diffusion Transformer for Video Generation

- Designed a transformer-based video diffusion model with 3D geometric consistency, integrating structure-from-motion camera trajectories, 2D Plücker embeddings, and epipolar attention for precise target pose conditioning, bridging video generation with 3D scene understanding.

Waymo

Software Engineer, Machine Learning Infrastructure, Intern.

Mountain View, CA

May. 2023 — Aug. 2023

- Proposed a differential learning rate (DLR) API that automatically divides model groups based on variable names, specified in the configuration files like proto, textproto, and dataclass.
- Created a modularized DLR training pipeline compatible with mainstream deep learning frameworks such as Tensorflow; Achieved a greater than 1.5% accuracy improvement across multiple tasks within the perception team.

ASUS Intelligent Cloud Services (AICS)

Machine Learning Engineer Intern.

Taipei, Taiwan

Jul. 2021 — Sept. 2021

- Operated Spark database to process and analyze more than 1,000,000 advertisements (Ads) and customer data logs, building an advertisement recommendation system to predict user browsing preferences with decision trees.
- Added over 300 features to solve model underfitting by organizing narrative categorization into multi-levels, improving the system's CTR (Click Through Rate) to 110% compared to previous performance.

PUBLICATIONS

* Equal contribution † Project lead

- [1] Yanru Qu*, **Cheng-Yen Hsieh***[†], Zaixiang Zheng, Ge Liu, Quanquan Gu.
“Protein Autoregressive Modeling via Multiscale Structure Generation”, *ICML 2026 (Oral, Top 0.7%)*.
- [2] **Cheng-Yen Hsieh***[†], Xinyou Wang*, Daiheng Z., Dongyu X., Fei YE, Shujian H., Zaixiang Z.[†], Quanquan Gu.
“Elucidating the Design Space of Multimodal Protein Language Models”, *ICML 2025 (Spotlight, Top 2.6%)*.
- [3] **Cheng-Yen Hsieh**, Kaihua Chen, Achal Dave, Tarasha Khurana, Deva Ramanan.
“TAO-Amodal: A Benchmark for Tracking Any Object Amodally”, *arXiv preprint arXiv:2312.12433*, 2023.
- [4] **Cheng-Yen Hsieh**, Chih-Jung Chang, Fu-En Yang, Yu-Chiang Frank Wang.
“Self-Supervised Pyramid Representation Learning for Multi-Label Visual Analysis and Beyond”, in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2023.
- [5] **Cheng-Yen Hsieh**, Yu-Chuan Chuang, An-Yeu Wu.
“C3-SL: Circular Convolution-Based Batch-Wise Compression for Communication-Efficient Split Learning”, in *2022 IEEE 32nd International Workshop on Machine Learning for Signal Processing (MLSP)*, 2022.
- [6] **Cheng-Yen Hsieh**, Yu-Chuan Chuang and An-Yeu Wu.
“FL-HDC: Hyperdimensional Computing Design for the Application of Federated Learning,” in *2021 IEEE 3rd International Conference on Artificial Intelligence Circuits and Systems (AICAS)*, 2021.

RESEARCH PROJECTS

Tracking Any Object Amodally ([Project Page](#))

Pittsburgh, PA

CMU Graduate Researcher | MSCV Capstone Project

Jan. 2023 — Mar. 2024

- Developed TAO-Amodal dataset to enhance amodal tracking, comprehending complete object structures from partial visibility, by including 17k objects spanning 880 categories and adapted evaluation metrics.
- Presented a light-weight plug-in module, amodal expander, to amodalize any existing tracker with limited training data; Introduced PasteNOcclude data augmentation technique to craft occlusion scenarios.
- Achieved significant 3.3% mAP and 1.6% mAP improvements on the detection and tracking of occluded objects, along with a >30% detection mAP boost for occluded people on the TAO-Amodal validation set.

Visual Question Answering with Vision Language Multi-Agent Debate

Pittsburgh, PA

MIT Research Assistant (Remote Collaboration)

Jul. 2023 — Mar. 2024

- Integrated vision language model agents into a collaborative debate setting to improve language responses through the synthesis of reasoning processes from multiple agents.
- Enhanced > 2% accuracy across diverse tasks such as instance counting, object attributes, and relation analysis.

Self-Supervised Representation Learning for Multi-Label Visual Tasks ([Paper](#), [GitHub](#))

Taipei, Taiwan

NTU Undergraduate Researcher

Aug. 2020 — Feb. 2022

- Designed an unsupervised visual pretext task for downstream multi-label classification tasks.
- Utilized multi-scale pyramid and patch-level correlation learning to leverage information across distinct scales.
- Outperformed the strongest and SOTA methods by 3.3% mAP and 2.5% mAP on COCO and VOC, respectively.

Split Learning for Efficient Image Recognition ([Paper](#), [GitHub](#))

Taipei, Taiwan

NTU Undergraduate Researcher

Jun. 2021 — Feb. 2022

- Proposed circular convolution-based batch-wise compression for split learning to compress multiple features into one by exploiting the quasi-orthogonality of high-dimensional features with circular convolution and superposition.
- Reduced 16x communication costs with negligible accuracy drops on CIFAR10 and CIFAR100, while improving 1152x memory and 2.25x computation overhead compared to the SOTA dimension-wise compression method.

Federated Learning for Efficient Image Recognition ([Paper](#), [GitHub](#))

Taipei, Taiwan

NTU Undergraduate Researcher

Jan. 2020 — Jun. 2021

- Designed computation-efficient FL-hyperdimensional computing (FL-HDC) to address high computation and communication burdens from previously training deep neural networks (DNN)s.
- Reduced communication costs by 23 times using bipolarization on model weights to cut communication costs, compensating for the precision loss of numerical values by proposing an adaptive retraining mechanism.

PATENT(S)

- [1] **Cheng-Yen Hsieh**, Xinyou Wang, Daiheng Zhang, Dongyu Xue, Fei Ye, Zaixiang Zheng, Quanquan Gu,
“Method, device and medium for protein language model”, US Patent App. 19/097,452, 2025.

MENTORSHIP & SERVICE

- [1] Mentored Yanru Qu (UIUC PhD) → PAR, ICML 2026 Oral; Serena Khoo (MIT EECS PhD), 2026.
- [2] Reviewer: NeurIPS 2026, ICML 2026, TMLR 2025–2026.